

AC Operating Conditions



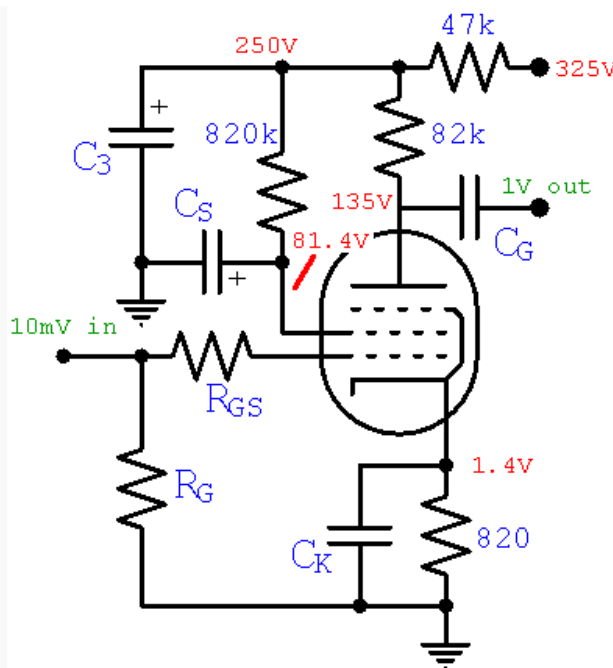
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Here is our design so far.



To eliminate negative feedback from cathode degeneration,¹ the bypass capacitor **CK** needs to be large enough to short all audio frequencies. The required value depends on the cathode impedance **ZK**, which is equal to the 820Ω cathode resistor **RK** in parallel with the reciprocal of the tube's transconductance **gm** that we estimated earlier:

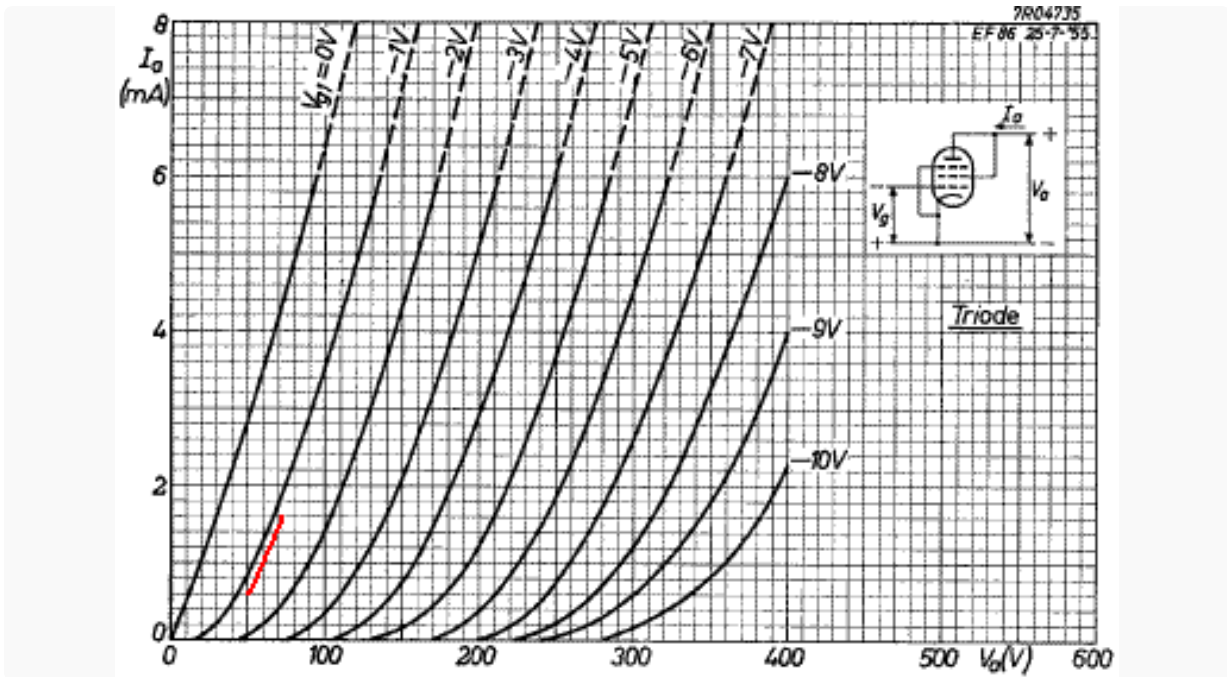
$$\mathbf{ZK = 820\Omega \parallel [1 / (1.5\text{ma/V})] = 368\Omega}$$

The lowest note on a guitar with standard tuning is 82 Hertz. So if we select a capacitor whose impedance is equal to the cathode impedance at a frequency 10 times lower, 8.2 Hertz, then we can be fairly certain that we've eliminated negative feedback:

$$\mathbf{CK = 1 / [2\pi(8.2\text{Hz})(368\Omega)] = 53\mu\text{F}}$$

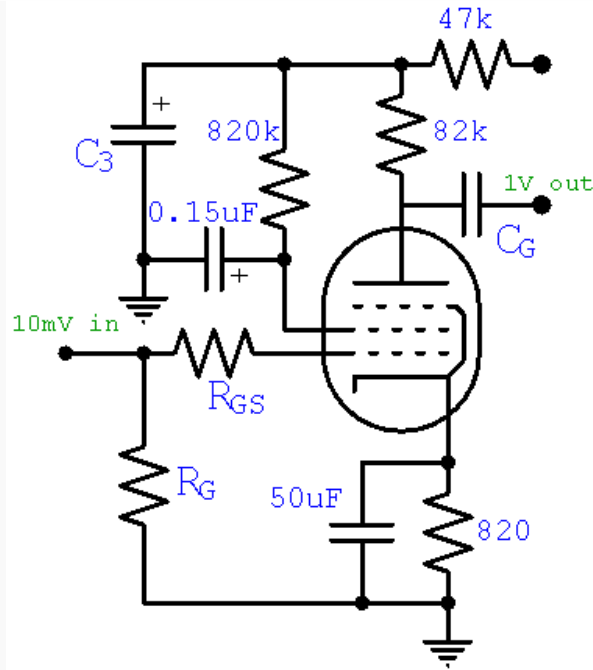
A 50μF value will suffice. Let's pull a similar trick to solve for the screen capacitor **CS**.

A pentode has screen resistance r_s that is analogous to the plate resistance r_p in a triode. Since the ratio of plate current to screen current is about 6 to 1, the screen resistance r_s is about 6 times greater than the plate resistance when the tube is triode-connected. Here are the triode-connected plate characteristics:



From the red line in the vicinity of the DC operating point, we observe that the plate voltage increases 22 volts when the plate current increases 1 milliamp. According to Ohm's Law this represents a plate resistance of 22kΩ, so screen resistance is about 6 times greater: 132kΩ. To make the impedance of the screen bypass capacitor C_s equal to 132kΩ at a tenth the lowest frequency of a guitar, the capacitor value is

$$C_s = 1 / [2\pi(8.2\text{Hz})(132\text{k}\Omega)] = 0.15\mu\text{F}$$



The Grid Circuit

The EF86 has about 5 picofarads of capacitance between the grid and cathode and only 0.05pF between the grid and plate. Because of the Miller Effect,² the capacitance between the grid and plate is multiplied by 1 plus the stage gain of 100, so the total Miller capacitance is

$$| \text{CM} = 5\text{pF} + (1 + 100)(0.05\text{pF}) = 10\text{pF}$$

A 12AX7 has ten times more parasitic capacitance, which is bad news for radio and why you won't find Marshall's favorite triode in the front end of a Philco. To ensure that radio-frequency suppression in our EF86 preamp begins at a cutoff frequency of 100kHz, the grid stopper resistor value needs to be

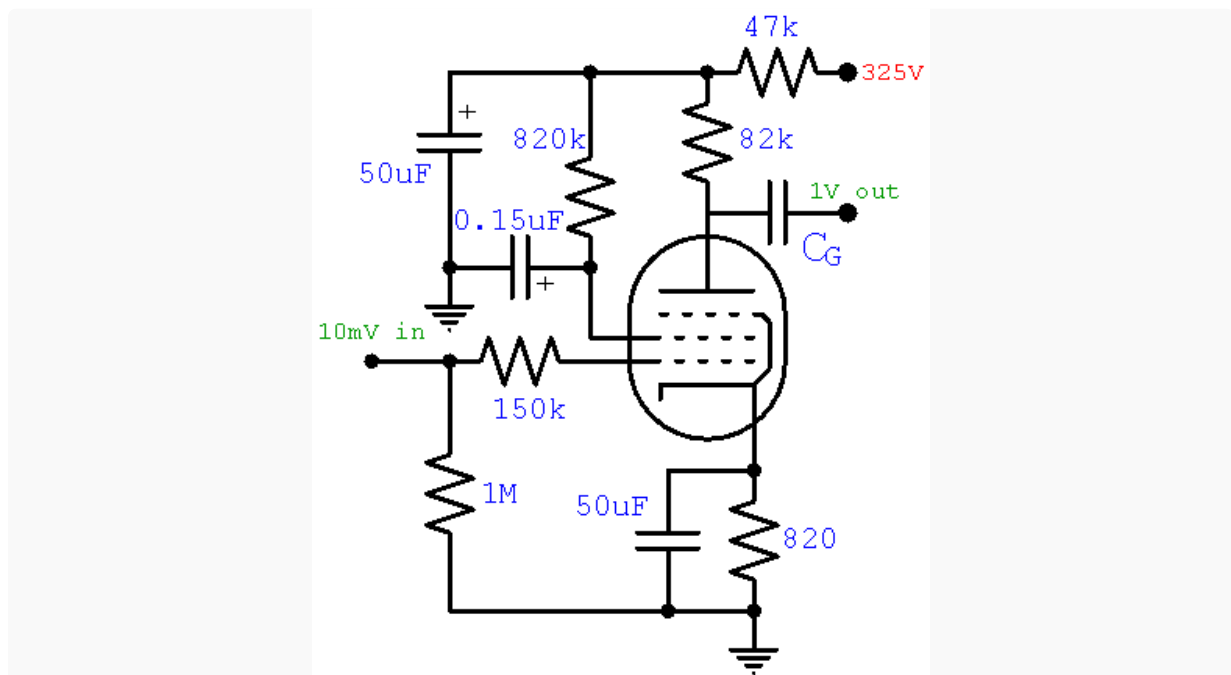
$$| \text{RGS} = 1 / [2\pi(100\text{kHz})(10\text{pF})] = 159\text{k}\Omega$$

We will use 150kΩ. A grid resistor value of **RG = 1MΩ** is a standard that places only a light load on the guitar circuit while providing sufficient grid leak.

As a final order of business, power supply filter capacitor **C3** needs to be large enough to mitigate AC ripple and decouple the preamp from later stages.³ The specific value depends on the rest of the amp. Since **R3** is a respectable 47k Ω , setting **C3** to 50 μ F creates an impressive 65dB of ripple suppression, as shown by this snapshot of our [RC Ripple Filter calculator](#).

ripple attenuation (dB)	-65.0
DC voltage drop (volts)	75.2
-3dB low-frequency cutoff (Hertz)	0.07
recalculate ↻	
resistor R (k Ω)	47
capacitor C (μ F)	50
AC power	☐ 50 Hertz
	☒ 60 Hertz
DC current (mA)	1.6

Here is the final design.



References

1Richard Kuehnel, [Guitar Amplifier Electronics: Basic Theory](#), (Seattle: Amp Books, 2018), p. 61.

2Richard Kuehnel, [Guitar Amplifier Electronics: Basic Theory](#), (Seattle: Amp Books, 2018), pp. 70-72.

3Richard Kuehnel, [Fundamentals of Guitar Amplifier System Design](#), (Seattle: Amp Books, 2019), pp. 100-113.